

1

- 6DOF MDP
- MDP
- Options SMDP

2 6DOF MDP

$$M_{LL} = (\mathcal{S}_{LL}, \mathcal{A}_{LL}, P_{LL}, R_{LL}, \gamma). \quad (1)$$

6DOF

$$s_t^{LL} = (\eta_t, \nu_t, g_t), \quad (2)$$

$$\eta_t = [x, y, z, \phi, \theta, \psi]^\top \quad \nu_t = [u, v, w, p, q, r]^\top \quad g_t \quad /$$

13

$$\mathcal{A}_{LL} = \{a_0, a_1, \dots, a_{12}\}. \quad (3)$$

surge/sway/heave yaw/pitch/roll

6DOF

$$\tau_t = f_{ctrl}(a_t, s_t^{LL}), \quad (4)$$

$$s_{t+1}^{LL} = f_{dyn}(s_t^{LL}, \tau_t, \xi_t), \quad (5)$$

ξ_t

/

$$r_t^{LL} = w_p \|e_p\| + w_o \|e_o\| + w_v \|\nu\| + w_e \|\tau\| + r_{goal} + r_{time}. \quad (6)$$

3 / MDP

$$M_H = (\mathcal{S}_H, \mathcal{A}_H, P_H, R_H, \gamma_H). \quad (7)$$

$$s_t^H = [p_p, v_p, p_e, v_e, \psi_e, b_p, b_e], \quad (8)$$

$$\mathcal{A}_H = \{ \quad / \quad / \quad \}. \quad (9)$$

$$s_{t+1}^H = F(s_t^H, a_t^H, \pi_{LL}). \quad (10)$$

$$r_t^H = \alpha \Delta d + \beta v_{close} + r_{catch} + r_{boundary} - r_{time}. \quad (11)$$

4 / Options SMDP

Options SMDP

$$M_S = (\mathcal{S}, \mathcal{O}, P_S, R_S, \gamma). \quad (12)$$

$$o = (\mathcal{I}_o, \pi_o, \beta_o), \quad (13)$$

$$P_S(s' \mid s, o) = \sum_{k \geq 1} P(s', k \mid s, o). \quad (14)$$

$$R_S(s, o) = \mathbb{E} \left[\sum_{i=0}^{k-1} \gamma^i r_{t+i} \right]. \quad (15)$$

5

6

6DOF 2D

6.1 6DOF

$$r_t^{LL} = r_{pos} + r_{ori} + r_{vel} + r_{ang} + r_{energy} + r_{att} + r_{stab} + r_{goal} + r_{time}. \quad (16)$$

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$$r_{pos} = w_{pos} \|e_p\| + \lambda_{imp} \Delta \|e_p\| + \lambda_{prox} \max(0, 1 - \|e_p\|) + \lambda_{dir} \langle \hat{e}_p, \hat{v} \rangle, \quad (17)$$

$$\Delta \|e_p\| \quad \langle \hat{e}_p, \hat{v} \rangle$$

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$$r_{ori} = w_{ori} \|e_o\|. \quad (18)$$

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$$r_{vel} = w_{vel} \max(0, \|\nu_{lin}\| - v_{lim}). \quad (19)$$

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$$r_{ang} = w_{ang} \max(0, \|\nu_{ang}\| - \omega_{lim}). \quad (20)$$

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$$r_{energy} = w_{energy} \frac{\sum_i |\tau_i|}{\tau_{max} \cdot n_{thr}}. \quad (21)$$

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$$r_{att} = r_{roll} + r_{pitch} + r_{rate} + r_{bonus}, \quad (22)$$

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$$\|e_p\| \quad \|e_o\|$$

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$$r_{time} < 0$$

reward weights/thresholds

6.2 2D

2D

$$r_t^{PE} = r_{improve} + r_{closing} + r_{align} + r_{boundary} + r_{time} + r_{hit} + r_{prox} + r_{catch}. \quad (23)$$

- $r_{improve} = w_{imp}(d_{t-1} - d_t)$
- $r_{closing} = w_{close} \text{clip}(v_{close}, -1, 1)$
- $r_{align} = w_{align} \langle \hat{v}_p, \hat{\Delta p} \rangle$
-
- $r_{time} = -\lambda_{time}$
- r_{hit}
- r_{prox}
- r_{catch}

6.3 2D 22

$$\Delta p = p_e - p_p, \Delta v = v_e - v_p, d = \|\Delta p\|,$$

$$v_{close} = -\frac{\Delta v \cdot \Delta p}{\|\Delta p\|} \quad (d > 0). \quad (24)$$

$$\text{pos_scale} = \frac{1}{\text{world_size}}, \quad \text{vel_scale} = \frac{1}{\max(v_{p_max}, v_{e_max})}, \quad \text{acc_scale} = \frac{1}{\max(a_{p_max}, a_{e_max})}. \quad (25)$$

22

- 1 : $p_{p,x} \cdot \text{pos_scale}$, 2 : $p_{p,y} \cdot \text{pos_scale}$,
- 3 : $v_{p,x} \cdot \text{vel_scale}$, 4 : $v_{p,y} \cdot \text{vel_scale}$,
- 5 : $\Delta p_x \cdot \text{pos_scale}$, 6 : $\Delta p_y \cdot \text{pos_scale}$,
- 7 : $\Delta v_x \cdot \text{vel_scale}$, 8 : $\Delta v_y \cdot \text{vel_scale}$,
- 9 : $d \cdot \text{pos_scale}$, 10 : $v_{close} \cdot \text{vel_scale}$,
- 11 : $\hat{v}_{e,x}$, 12 : $\hat{v}_{e,y}$,
- 13 : $(\text{world_size} - p_{p,y}) \cdot \text{pos_scale}$,
- 14 : $(p_{p,y} + \text{world_size}) \cdot \text{pos_scale}$,
- 15 : $(p_{p,x} + \text{world_size}) \cdot \text{pos_scale}$,
- 16 : $(\text{world_size} - p_{p,x}) \cdot \text{pos_scale}$,
- 17 : $\min(\text{world_size} - |p_{e,x}|, \text{world_size} - |p_{e,y}|) \cdot \text{pos_scale}$,
- 18 : $\|v_p\| \cdot \text{vel_scale}$, 19 : $\|v_e\| \cdot \text{vel_scale}$,
- 20 : $\hat{v}_p \cdot \hat{\Delta p}$,
- 21 : $a_{p,x} \cdot \text{acc_scale}$, 22 : $a_{p,y} \cdot \text{acc_scale}$.

6.4 6DOF

$$R(\phi, \theta, \psi)$$

$$e_p^{body} = R(\phi, \theta, \psi)^\top (g_p - \eta_{pos}). \quad (26)$$

$$e_o = \text{wrap}(\eta_{ori} - g_{ori}) \in [-\pi, \pi]^3. \quad (27)$$

- $e_p^{body}/10$
- e_o/π
- $[u, v, w]/2$
- $[p, q, r]/2$
- $e_p^{body}/10$
- g_{ori}/π
- $e_p^{body}/10 \quad e_o/\pi$

7

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cd docs
pdflatex markov_modeling_cn.tex
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